

Find recent machine learning papers that discuss 'formatting fragility', 'instruction forgetting', or 'format collapse' in Large Language Models (LLMs) when tested on Out-of-Distribution (OOD) real-world data after being trained or fine-tuned on synthetic data. Specifically, look for papers that differentiate between an LLM's underlying reasoning accuracy remaining intact, while its ability to follow strict formatting constraints fails.

Recent Papers on Format Fragility in Synthetic-Data LLM Tuning

The literature separates this topic into **formatting failures**, **instruction forgetting**, and **mitigation under distribution shift**. The closest matches are papers showing that synthetic or skewed fine-tuning data can preserve benchmark or reasoning competence while degrading instruction-following, formatting compliance, or output diversity on OOD or real-world inputs (Chen et al., 2024).

Direct Matches

These papers most directly match the requested failure mode: the model's **underlying capability is partly preserved**, but its **format or instruction compliance breaks**.

Paper	Synthetic/OOD Setting	Core Dissociation	Relevance
Chen et al., 2024, <i>Unveiling the Flaws</i>	Synthetic Q-A fine-tuning	Uniform synthetic formats cause pattern overfitting and output-distribution shift that reduces instruction-following, while benchmark ability is preserved by mitigation	Very high (Chen et al., 2024)
Yun et al., 2025, <i>The Price of Format</i>	Structured prompt fine-tuning with format mismatch at inference	Format consistency matters for structure-sensitive tasks, but much less for knowledge-heavy tasks	Very high (Yun et al., 2025)
Lim et al., 2025, <i>Format Inertia</i>	Real-world medical dialogue with skewed training distribution	Outputs remain format-correct but become repetitive and diagnostically uninformative in underrepresented long dialogues	Very high (Lim et al., 2025)
Chen et al., 2024, <i>CoIN</i>	Continual instruction tuning under task shift	Forgetting is driven more by intention alignment failure than knowledge loss	High (Chen et al., 2024)
Jiang et al., 2024, <i>Instruction Vector</i>	Fine-tuning under forgetting	Fine-tuning appears to add specialized patterns rather than erase prior skills, making forgetting look like lost instruction following	High (Jiang et al., 2024)

FIGURE 1 Most direct papers linking synthetic or shifted fine-tuning to format or instruction failures distinct from core capability loss.

Formatting Failures

Synthetic data with **uniform Q-A structure** can induce pattern overfitting and shift the output distribution, which reduces instruction-following rather than simply lowering benchmark scores (Chen et al., 2024). An unlearning-based correction can restore instruction-following while preserving foundational benchmark performance, which is strong evidence for a compliance-versus-capability split (Chen et al., 2024).

Structured templates and special tokens also create a broader **format dependence**. Format consistency between fine-tuning and inference is crucial on structure-sensitive tasks like GSM8K and IFEval, but has only marginal influence on knowledge-heavy tasks like MMLU and WebQuestions, implying that formatting can fail before knowledge does (Yun et al., 2025).

- **Diversity collapse** persists even at high temperature under structured formatting (Yun et al., 2025).
- **Structural tokens** constrain output space more than content-heavy knowledge tasks (Yun et al., 2025).
- Diffusion-based synthetic generation was proposed partly because prompted LLM synthesis struggles with **structured formatted data** (Zhou et al., 2024).

Instruction Forgetting

Several papers frame the same phenomenon as **instruction forgetting** or catastrophic forgetting after fine-tuning. Continual instruction tuning in multimodal LLMs shows that failure in intention alignment is the main source of forgetting, not knowledge preservation for reasoning (Chen et al., 2024). Instruction-vector analysis likewise argues that fine-tuning often adds specialized reasoning patterns instead of erasing previous skills, so the observed failure can look like lost instruction following more than destroyed capability (Jiang et al., 2024).

OOD reasoning can also **decline late in SFT** even while ID loss falls and ID accuracy rises, showing specialization can mask broader capability loss (Jin et al., 2025). RL then recovers some lost OOD ability, but does not exceed the model's earlier OOD peak or create fundamentally new reasoning skills (Jin et al., 2025).

- General fine-tuning often degrades **general-domain tasks** after specialization (Qiu et al., 2026).
- Distribution shift during fine-tuning reduces **general capability, safety, and helpfulness** (Yang et al., 2024).
- Synthetic fine-tuning can reduce non-target degradation when generated sequences contain **lower-perplexity tokens** (Wu et al., 2025).

OOD Real-World Manifestations

The clearest real-world analogue is **Format Inertia** in medical pre-consultation. When long dialogues are underrepresented in training, the model falls back to repetitive question templates that satisfy superficial format constraints but fail to gather new diagnostic information (Lim et al., 2025). The model preserves surface form yet cannot integrate long-range context effectively, which is very close to the reasoning-intact / formatting-fails distinction you asked for (Lim et al., 2025).

Recursive synthetic training also shows **collapse can be format-conditional rather than uniform**. Short-answer prompts remain stable longest, while few-shot formats collapse fastest, indicating that degradation timing depends on instructional format itself (Keisha et al., 2025). In parallel, models can remain fluent while factual reliability declines, showing another dissociation between surface competence and underlying correctness (Keisha et al., 2025).

- Recursive synthetic training narrows the **output distribution** and reduces ability to model human text (Schaffelder & Gatt, 2025).
- Syntactic diversity declines the most under self-training, more than semantic diversity (Schaffelder & Gatt, 2025).
- Domain-specific synthetic corpora delay collapse relative to general recursive training (Keisha et al., 2025).

Overall, the strongest matches are **Chen et al. 2024**, **Yun et al. 2025**, **Lim et al. 2025**, **Chen et al. 2024 CoIN**, and **Jiang et al. 2024**. Together, they support the conclusion that after synthetic or shifted fine-tuning, LLMs often retain substantial underlying competence while **format adherence, instruction alignment, or structurally constrained output behavior** fails first.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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